

WEEK 2 DAY 14

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Battery SOC Estimation (EDA & Feature Engineering)

Dataset Architecture & Ingestion

- **Objective:** Consolidate heterogeneous battery cycle CSV logs into a standardized DataFrame.
- **Key Steps & Specs:**
- **Directory Parsing:** Standardized loading across 36 separate battery test files (e.g., C20DisCh, HPPC, HWFET, LA92, UDDS, US06, Mixed1–8).
- **Downsampling:** Applied `SAMPLE_EVERY_N = 10` downsampling for scalable model training.
- **Header Alignment:** Dynamic line matching on "Time Stamp,Step,Status" to skip variable-length file metadata headers.
- **Battery Parameter:** Cell Rated Capacity set to 3.0 Ah(*RATED_CAPACITY_AH*).

Data Schema & Cleaning Verification

- **Clean Dataset Shape:** 104,560 rows × 17 columns

Variable Category	Column Names	Data Types
Metadata	timestamp, step, status, procedure, test_type, file_id	object / float64
Operational Metrics	prog_time, step_time, cycle, cycle_level, cnt	object / float64
Sensor Readings	voltage_v, current_a, cell_temp_c, capacity_ah, wh_accu	Float64/IPYNB
Environmental	ambient_temp_c (25°C)	Int64/IPYNB

Data Quality Audit:

Missing Values: 0 missing values across all sensor features.
Duplicates: 0 duplicate rows after step-transition boundary filtering.

Operational Status & Drive Cycle Distribution

- **Status Breakdown (Logger States):**

- TABLE (Dynamic Drive Profiles): 85,516 rows
- PAU (Pause / Rest Periods): 16,989 rows
- CHA (Charge Cycles): 1,335 rows
- DCH (Discharge Cycles): 719 rows

- **Top Drive Cycle Datasets:**

- **UDDS:** 15,965 rows | **LA92:** 10,077 rows | **Mixed8:** 8,592 rows
- **US06:** 4,016 rows | **HWFET:** 601 rows | **HPPC:** 4,805 rows

Feature Engineering & Target Formulation

- **1. Target Feature (soc_percent):**

- $$\text{SOC (\%)} = 100 \times \left(1 + \frac{\text{capacity_ah}}{\text{RATED_CAPACITY_AH}} \right)$$

- Clipped to range $[0,100]\%$ to prevent out-of-bound edge cases.

- **2. Instantaneous Power (power_w):**

- $$\text{Power (W)} = \text{voltage_v} \times \text{current_a}$$

- Captures energy transfer dynamics during charging (> 0) and discharging (< 0).

Target Distribution Analysis

- **Engineered Target Statistics:**

- **Mean:** 53.30% | **Standard Deviation:** 24.71%
- **Min / Max:** 0.00% / 100.00%
- **25th Percentile:** 34.78% | **50th (Median):** 52.70% | **75th Percentile:** 74.52%

- **Key Observations:**

- Balanced distribution across full SOC spectrum (0 — 100%).
- High coverage in standard EV operating range (20% — 80%), ideal for machine learning model training.

Summary & Next Steps for Day 15

- **Summary of Progress:**

- Successfully ingested and cleaned 36 files at 25°C ambient temperature.
- Created robust continuous target variable (soc_percent) and power feature (power_w).

- **Day 15 Execution Plan:**

- Feature correlation matrix & multicollinearity check (V, I, T, P).
- Train/Test split strategies (Drive-cycle based split vs. random split).
- Baseline regression modeling (Random Forest / XGBoost / Neural Networks).

THANK YOU